Lab Worksheet 2: Conditionals

CORE 103 Computational Thinking

Fall 2026

Instructions

- This LW must be submitted online via CANVAS. You are required to submit a zip file that contains all the *.py* files.
- The zip file should be named as *LW2_aa123456.zip* where *aa123456* will be replaced with your student id.
- **Files that don't follow the appropriate naming convention will not be graded.**

Exercises

1. Movie Night Budget

This question should be implemented in the file ***q1.py****.*

Four university friends are planning a movie night after a long lab day. They have saved a total of Rs. 10000 for the tickets, so they first check the price before confirming the plan in the group chat.

The cinema website shows that one ticket costs `p` rupees. Since all four friends need tickets, decide whether their plan fits the money they have. If it fits, print "YES"; otherwise, print "NO".

Constraints

- $\texttt{p} \in \mathbb{N}$
- $100 \leq \texttt{p} \leq 3000$

Interaction

The input contains a single integer: `p`.

The output must be "YES" if the four friends can afford the tickets, and "NO" otherwise.

Sample

Input	Output
1000	YES
3000	NO

In the first case, p=1000. Four tickets cost $1000 \times 4 = 4000$, which is within the budget, so the output is "YES".

In the second case, p=3000. Four tickets cost $3000 \times 4 = 12000$, which is above the budget, so the output is "NO".

Exercise

Indicate the outputs for the given inputs in **q1.py**.

Input	Output
2500	
2800	
2600	

Python Code

Implement the python code in **q1.py**.

2. Campus Direction Game

*This question should be implemented in the file **q2.py**.*

During orientation week, new students are playing a campus direction game. One student starts by facing North near the main gate. Every minute, the game host asks the student to turn 90° clockwise.

The student can only face North, East, South, or West. Given the number of minutes `m`, determine the direction the student is facing at the end.

Constraints

- $\texttt{m} \in \mathbb{N}$
- $1 \leq \texttt{m} \leq 360$

Interaction

The input contains a single integer: `m`.

The output must be one of "North", "East", "South", or "West".

Sample

Input	Output
1	East
6	South

In the first case, `m`=1. After one clockwise turn, the student faces East.

In the second case, `m`=6. Six clockwise turns leave the student facing South.

Exercise

Indicate the outputs for the given inputs in **q2.py**.

Input	Output
8	
15	
253	

Python Code

Implement the python code in **q2.py**.

3. Chess Club Rooks

*This question should be implemented in the file **q3.py**.*

During chess club practice, two students place rooks on a standard 8×8 chessboard to test attacking positions. Rows and columns are numbered from 1 to 8.

The first rook is at row `r1`, column `c1`, and the second rook is at row `r2`, column `c2`. Determine whether either rook can attack the other in one straight move.

Constraints

- $\texttt{r1}, \texttt{c1}, \texttt{r2}, \texttt{c2} \in \mathbb{N}$
- $1 \leq \texttt{r1}, \texttt{c1}, \texttt{r2}, \texttt{c2} \leq 8$
- The two rooks are on different squares.

Interaction

The input contains four space-separated integers: `r1`, `c1`, `r2`, and `c2`.

The output must be "Yes" if the rooks attack each other, and "No" otherwise.

Sample

Input	Output
1 2 5 2	Yes
1 1 8 8	No

In the first case, both rooks are in column 2, so they attack each other.

In the second case, the rooks are not in the same row or the same column, so they do not attack each other.

Exercise

Indicate the outputs for the given inputs in **q3.py**.

Input	Output
1 2 1 5	
2 4 4 8	
3 3 4 3	

Python Code

Implement the python code in **q3.py**.

4. Weekend Plan Boost

*This question should be implemented in the file **q4.py**.*

After a long week of classes, a student wants to plan a short weekend hangout with friends. There are three possible plans: a cafe visit, a movie night, and a gaming session. The fun scores of these plans are `A`, `B`, and `C`, respectively.

Because everyone has assignments due, the group only has time for exactly two of the three plans. Given `A`, `B`, and `C`, determine the maximum total fun score the group can get from choosing two plans.

Constraints

- $A, B, C \in \mathbb{N}$
- $1 \leq A, B, C \leq 10^8$

Interaction

The input comprises a single line containing 3 space separated integers denoting the values of A, B, C respectively.

The output should contain a single integer denoting the maximum total fun score.

Sample

Input	Output
4 2 8	12
10 10 100	110

In the first case, $(A, B, C) = (4, 2, 8)$. The best two plans have scores 8 and 4, so the maximum total fun score is $8 + 4 = 12$.

In the second case, $(A, B, C) = (10, 10, 100)$. The best two plans have scores 100 and 10, so the maximum total fun score is $100 + 10 = 110$.

Exercise

Indicate the outputs for the given inputs in **q4.py**.

Input	Output
18 271 31	
127 2933 182	
21 2 18	

Python Code

Implement the python code in **q4.py**.

5. Lab Score Rounding

*This question should be implemented in the file **q5.py**.*

A first-year programming lab uses an automatic grading portal. The portal follows a score-rounding policy before displaying the final lab score.

For a score g, the policy is:

- Scores below 38 are not rounded because they remain failing scores.
- Otherwise, if the difference between g and the next multiple of 5 is less than 3, the score is rounded up to that multiple of 5.
- If the difference is not less than 3, the score remains unchanged.

Given the original score g, determine the final score shown by the portal.

Constraints

- $\mathsf{g} \in \mathbb{Z}_{\geq 0}$
- $0 \leq \mathsf{g} \leq 100$

Interaction

The input contains a single integer: g.

The output must be the final score after applying the rounding policy.

Sample

Input	Output
84	85
57	57

In the first case, the next multiple of 5 after 84 is 85. The difference is 1, which is less than 3, so the score becomes 85.

In the second case, the next multiple of 5 after 57 is 60. The difference is 3, which is not less than 3, so the score remains 57.

Exercise

Indicate the outputs for the given inputs in **q5.py**.

Input	Output
29	
78	
51	

Python Code

Implement the Python code in **q5.py**.

6. Packing Boxes

*This question should be implemented in the file **q6.py**.*

A student is moving three hostel supply boxes. The boxes are stacked one above the other, with weights `w1`, `w2`, and `w3` from top to bottom. The student can carry at most `s` kg in one trip.

In each trip, the student may carry one or more consecutive boxes from the top of the current stack, provided their total weight does not exceed `s`. Before each trip, the student may reverse the order of the remaining stack if it helps. Find the minimum number of trips needed to move all three boxes.

Constraints

- $\texttt{w1}, \texttt{w2}, \texttt{w3}, \texttt{s} \in \mathbb{N}$
- $1 \leq \texttt{w1}, \texttt{w2}, \texttt{w3} \leq \texttt{s}$
- $1 \leq \texttt{s} \leq 50$

Interaction

The input contains four space-separated integers: `w1`, `w2`, `w3`, and `s`.

The output must be the minimum number of trips needed.

Sample

Input	Output
8 4 6 10	2
7 6 5 10	3

In the first case, the total weight is $8 + 4 + 6 = 18$, which is greater than the capacity 10. However, the student may reverse the stack so that the boxes of weights 6 and 4 are on top. These two boxes have total weight 10, so they can be carried together in one trip, and the remaining box of weight 8 is carried in the second trip. Therefore, 2 trips are needed.

In the second case, the total weight is also $7 + 6 + 5 = 18$, with the same capacity 10. However, neither adjacent pair fits: $7 + 6 = 13$ and $6 + 5 = 11$. Reversing the stack does not change these adjacent pair sums, so no two boxes can be carried together. Therefore, 3 trips are needed.

Exercise

Indicate the outputs for the given inputs in **q6.py**.

Input	Output
2 1 3 6	
8 5 4 10	
6 5 6 10	

Python Code

Implement the python code in **q6.py**.

7. DEBUG: Favorite Number Game

*This question should be implemented in the file **q7.py**.*

Three friends are playing a number game in the common room. Alice has a special liking for numbers connected to 7, Bob has a special liking for numbers connected to 9, and Charlie takes any number that neither of them claims.

Alice claims a number only if it is even and also divisible by 7. Bob claims a number only if it is odd and also divisible by 9. Given A, determine which friend gets the number.

Constraints

- $A \in \mathbb{N}$
- $1 \leq A \leq 1000$

Interaction

The input comprises a single line containing an integer denoting the value of A.

The output must contain the name of the friend who gets the number. The name shall be one of the following: Alice, Bob, or Charlie.

Sample

Input	Output
7	Charlie
27	Bob

In the first case, 7 is not even, so Alice does not accept it. It is odd, but it is not a multiple of 9, so Bob does not accept it either. Therefore, Charlie gets the number.

In the second case, 27 is odd and a multiple of 9. Therefore, Bob accepts it and gets the number.

Exercise

Indicate the outputs for the given inputs in **q7.py**.

Input	Output
14	
21	
63	

Debugging Task

The file **q7.py** contains a program intended to solve the Favorite Number Game. However, the program contains logical errors.

Run the program using the provided test cases, identify the logical errors, and correct the code so that it produces the expected outputs.